

Chapter - Two

Free Electron theory of Conduction in metal.

Crystalline structure $\Rightarrow$

Following are the three crystalline structures

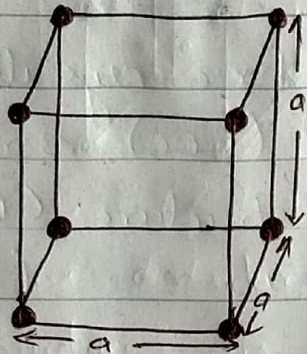
- (i) Simple cubic crystal (SCC) structure.
- (ii) Body Centred cubic structure (BCC)
- (iii) Face " " " (FCC)

(i) Simple cubic structure $\Rightarrow$

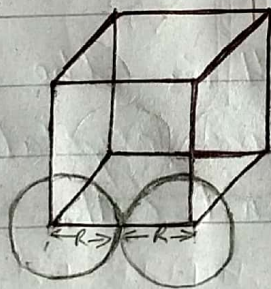
Before studying the crystalline structure we need to be clear about the following terms.

- (a) unit cell $\Rightarrow$ A unit cell is the smallest representation of an entire crystal. All crystals ^{lattices} are built of repeating unit cells.
- (b) Co-ordination Number $\Rightarrow$ In a unit cell, an atom's co-ordination number is the number of atoms it is touching.
- (c) Atomic ~~any~~ radius $\Rightarrow$ It is half of the distance ~~&~~ between the neighbouring atoms in a crystal.

In a simple cubic crystal, the structure is as follows.



Consider a cube of ~~radius 'r'~~ side 'a' and in SCC structure, an atom is present at each corner.



Atomic radius (R)

Here, from fig:

$$a = 2R$$

$$\therefore R = \frac{a}{2}$$

$$\text{No. of atoms per unit cell} = \frac{1}{8} \times 8 = 1$$

$$\text{Co-ordination number} = 6$$

$$\text{Volume of a unit cell} = a^3$$

$$\text{Volume of an atom} = \frac{4}{3} \pi R^3$$

$$= \frac{4}{3} \pi \left(\frac{a}{2} \right)^3 = \frac{4}{3} \pi a^3 \times \frac{1}{8}$$

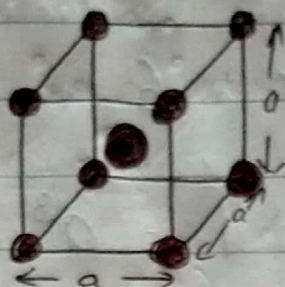
$$= \frac{1}{6} \pi a^3$$

$$\therefore \text{Packing density} = \frac{\text{No. of atoms per unit cell} \times \text{Volume of an atom}}{\text{Volume of a unit cell}}$$

$$= \frac{1 \times \frac{1}{6} \pi a^3}{a^3}$$

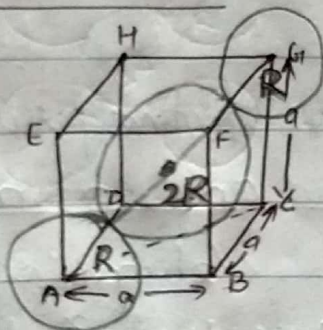
$$= 0.52$$

(ii) Body Centered Cubic structure



Here, also we consider a cube of side 'a'.
In BCC structure, an atom is present at the centre of the body and an atom is present at the each corner of the cube.

Atomic Radius (R)



$$AG = R + 2R + R = 4R$$

$$AC = \sqrt{AB^2 + BC^2} = \sqrt{a^2 + a^2} = \sqrt{2a^2} = \sqrt{2}a$$

$$GC = a$$

According to pythagorus theorem,

$$AG^2 = AC^2 + GC^2$$

$$(4R)^2 = 2a^2 + a^2$$

$$\text{or } 16R^2 = 3a^2$$

$$\text{or } R^2 = \frac{3a^2}{16}$$

$$\therefore R = \frac{\sqrt{3}a}{4}$$

Now, Co-ordination - number = 8

$$\text{No. of atoms per unit cell} = 1 + \frac{1}{8} \times 8 = 2$$

$$\text{Volume of a unit cell} = a^3$$

$$\text{Volume of an atom} = \frac{4}{3} \pi R^3$$

$$= \frac{4}{3} \pi \left(\frac{\sqrt{3}a}{4} \right)^3$$

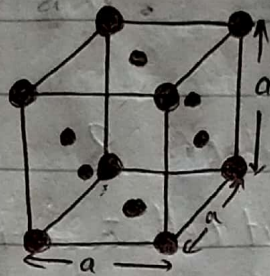
$$\text{Packing density} = \frac{\text{No. of atoms per unit cell} \times \text{Volume of an atom}}{\text{Volume of a unit cell}}$$

$$= \frac{2 \times \frac{4}{3} \pi \left(\frac{\sqrt{3}a}{4} \right)^3}{a^3}$$

$$= 0.6801$$

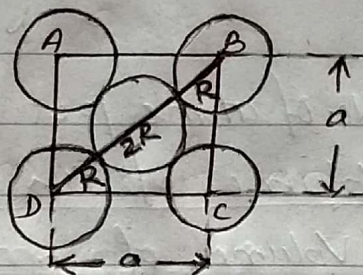
(iii) Face centered cubic structure $\Rightarrow$

Here also, we consider a cube of side 'a'. In FCC structure, an atom is present at each faces of the cube and an atom is present at the each corner of the cube.



Atomic Radius (R)

Let us take one face now, In each face, one atom is present at the face and 4 atoms at each corner.



In the figure,

$$BD = R + 2R + R = 4R$$

According to pythagoras theorem,

$$BD^2 = CD^2 + BC^2$$

$$\text{or } (4R)^2 = a^2 + a^2$$

$$\text{or } 16R^2 = 2a^2$$

$$\text{or } R^2 = \frac{2a^2}{16}$$

$$\therefore R = \frac{a}{2\sqrt{2}}$$

$$\text{Co-ordination number} = \frac{1}{8} \times 8 + 12$$

$$\text{No. of atoms per unit cell} = 1 \times \frac{1}{8} + 6 \times \frac{1}{2}$$
$$= 4$$

$$\text{Volume of a unit cell} = a^3$$

$$\text{Volume of an atom} = \frac{4}{3} \pi R^3$$
$$= \frac{4}{3} \pi \left(\frac{a}{2\sqrt{2}} \right)^3$$

$$\text{Packing density} = \frac{\text{No. of atoms per unit cell} \times \text{Volume of an atom}}{\text{Volume of a unit cell}}$$

$$= \frac{4 \times \frac{4}{3} \pi \left(\frac{a}{2\sqrt{2}} \right)^3}{a^3}$$
$$= 0.74$$

Lattice constant $\Rightarrow$

Lattice constant or lattice parameter refers to the physical dimension of unit cells in a crystal lattice. In case of cubic crystal structure, lattice constants are equal and we refer to 'a'.

Question $\Rightarrow$ Calculate lattice constants, face diagonal, body diagonal and packing density of SCC, BCC and FCC unit cells.

Effective mass of an electron $\Rightarrow$

A particle's effective mass is the mass that it seems to have when responding to forces or the mass that it seems to have with other particles in thermal distribution.

Kinetic energy of free particle is given as:

$$E = \frac{1}{2}mv^2$$
$$= \frac{m^2v^2}{2m}$$

$$\text{or, } E = \frac{p^2}{2m}$$

$$\left[\because p = \hbar k \text{ where, } \hbar = \frac{h}{2\pi} \right]$$

$$\text{or } E = \frac{\hbar^2 k^2}{2m}$$

$$\text{and } k = \frac{2\pi}{\lambda}$$

differentiating above eqⁿ w.r.t. k then,

$$\frac{\partial E}{\partial k} = \frac{\hbar^2 k}{m}$$

$$\text{or } \frac{1}{\hbar} \frac{\partial E}{\partial k} = \frac{\hbar k}{m} \quad \text{--- (i)}$$

$$\text{Also, } \lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\text{or } v = \frac{h}{m\lambda}$$

$$\text{or } v = \frac{h}{2\pi} \times \frac{2\pi}{\lambda} \times \frac{1}{m}$$

$$\therefore v = \frac{\hbar k}{m} \quad \text{--- (ii)}$$

From (i) & (ii) we get,

$$\frac{1}{\hbar} \frac{\partial E}{\partial k} = v \quad \text{--- (iii)}$$

Also, acceleration of particle

$$a = \frac{\partial v}{\partial t}$$

$$\therefore a = \frac{1}{\hbar} \frac{\partial}{\partial t} \frac{\partial E}{\partial K} \quad [\text{using relation (iii)}]$$

$$\frac{\partial v}{\partial t} = \frac{1}{\hbar} \frac{\partial}{\partial t} \cdot \frac{\partial E}{\partial K}$$

$$\frac{\partial}{\partial t} \left(\frac{\hbar K}{m} \right) = \frac{1}{\hbar} \left\{ \frac{\partial}{\partial K} \cdot \frac{\partial K}{\partial t} \cdot \frac{\partial E}{\partial K} \right\}$$

$$\text{or, } \frac{\hbar}{m} \frac{\partial K}{\partial t} = \frac{1}{\hbar} \cdot \frac{\partial K}{\partial t} \cdot \frac{\partial^2 E}{\partial K^2}$$

$$\therefore \frac{\hbar^2}{m} = \frac{\partial^2 E}{\partial K^2}$$

$$\therefore m = \frac{\hbar^2}{\frac{\partial^2 E}{\partial K^2}}$$

So, mass depends upon curvature of energy in K -space. (E - K diagram).

For solid, the above expression of mass is known as effective mass denoted by m^* .

$$\therefore m^* = \frac{\hbar^2}{\left(\frac{\partial^2 E}{\partial K^2} \right)}$$

4) Effective mass of electron is positive for $\frac{\partial^2 E}{\partial k^2} > 0$ i.e.

Curvature is Concave upward.

4) Effective mass of electron is negative for $\frac{\partial^2 E}{\partial k^2} < 0$, i.e. Curvature is Concave downwards.

4) For certain values of k the curvature is zero pointing towards the infinite mass.

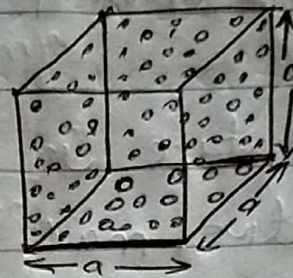
4) mass is positive and electron is accelerated in the direction of applied external electric field.

4) mass is negative and electron is decelerated due to the external applied field.

Thermal velocity of electron at equilibrium $\Rightarrow$

For N molecules of gas in a container, each molecule has its random velocity, change in momentum following collision with the wall of container.

Change in momentum of a particle along +ve x-direction after suffering collision is,



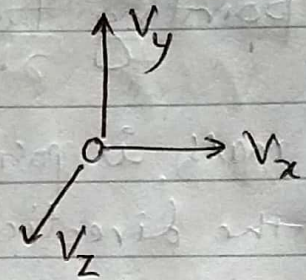
$$\Delta P_x = 2mV_x$$

Fig: Gas molecules in a container.

[$\because V_x$ is the velocity of gas molecules in the +ve x-direction]

$m \Rightarrow$ mass of a gas molecule

$$\text{Time interval, } \Delta t = \frac{2a}{V_x}$$



$$\text{Force on wall} = \frac{\Delta P_x}{\Delta t}$$

$$= \frac{2mV_x}{\frac{2a}{V_x}}$$

$$= \frac{mV_x^2}{a}$$

$$= \frac{mV_x^2}{a}$$

$$\text{Pressure on wall (P)} = \frac{\text{Total force due to N-particle}}{\text{Area of face}}$$

$$= \frac{mV_{x_1}^2 + mV_{x_2}^2 + mV_{x_3}^2 + \dots + mV_{x_n}^2}{a(a^2)}$$

✓

$$P = \frac{mN\bar{V}_x^2}{a^3} \left[\because \bar{V}_x^2 = \frac{V_{x1}^2 + V_{x2}^2 + \dots + V_{xn}^2}{N} \right]$$

$$= \frac{Nm\bar{V}_x^2}{a^3}$$

Since motion is in 3-D

$$\bar{V}^2 = \bar{V}_x^2 + \bar{V}_y^2 + \bar{V}_z^2$$

Assuming $\bar{V}_x = \bar{V}_y = \bar{V}_z$

$$\bar{V}^2 = 3\bar{V}_x^2$$

$$\bar{V}_x^2 = \frac{\bar{V}^2}{3}$$

$\left[\bar{V}^2 = \text{mean square velocity of molecule} \right]$

$$\therefore P = \frac{Nm\bar{V}^2}{3a^3}$$

$$\text{or } P = \frac{Nm\bar{V}^2}{3V} \quad \left[\text{where, } V = \text{Volume of container which is a cube} \right]$$

$$\text{or } PV = \frac{Nm\bar{V}^2}{3}$$

Average thermal energy per degree of freedom is $\frac{KT}{2}$. For 3-degree of freedom, it is $\frac{3KT}{2}$.

Kinetic energy (K.E) per molecule is $\frac{m\bar{V}^2}{2}$

$$\text{So, } K.E = \frac{1}{2}m\bar{V}^2 = \frac{3KT}{2}$$

$\therefore V_{th} = \sqrt{\frac{3kT}{m}}$, where, V_{th} is mean thermal velocity.

Electron Mobility, Conductivity & Resistivity $\Rightarrow$

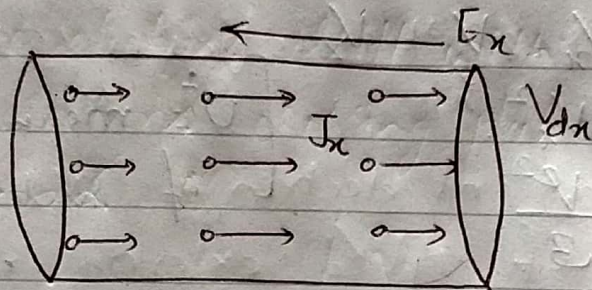


Fig: Electron drifting on application of Electric field.

Due to random thermal motion no net current is detected, so to maintain the flow of electron electric field is applied. If V is applied field across length l then the electric field is given by,

$$E = \frac{V}{l} \quad (i)$$

In this field, electron experiences force equal to

$$F = -eE \quad \text{--- (1)}$$

Also we know,

$$F = m^* a$$

where, m^* = effective mass of electron

a = acceleration of the electron

then,

$$a = \frac{F}{m^*} = \frac{-eE}{m^*} \quad \text{--- (2)}$$

Due to the electric field, electron acquires velocity in opposite direction to applied field called drift velocity.

The average time between two collision is termed as mean free time or relaxation time denoted by τ .

then,

$$V_{avg} = a\tau$$

Also, the average velocity (V_{avg}) is called drift velocity. So, neglecting -ve sign, we get.

(Since electron drift in opposite direction of applied field)

$$V_d = a \tau$$

$$V_d = \left(\frac{eE}{m^*} \right) \tau$$

$$\therefore V_d = \mu_d E$$

Thus, $\mu_d = \frac{e\tau}{m}$, is called as drift mobility.

If all the electrons drift with velocity V_d , then no. of electrons crossing plane of unit area per second is

$$\begin{aligned} \text{Electron flow} &= \frac{\text{No. of electrons}}{\text{Area} \times \text{time}} \\ &= \frac{\text{No. of electrons} \times \text{length}}{\text{Area} \times \text{length} \times \text{time}} \\ &= \frac{\text{No. of electrons}}{\text{Area} \times \text{length}} \times \frac{\text{length}}{\text{time}} \\ &= N_e \times \text{Velocity} \\ &= N_e \times V_d \end{aligned}$$

where, N_e = electron density.

$$\begin{aligned}\text{Current density (J)} &= \text{charge} \times \overset{\text{electron}}{\text{flow}} \\ &= e \times N_e \times V_d \\ &= e N_e V_d\end{aligned}$$

Also, we know,

$$V_d = \mu_d E$$

$$\therefore J = e N_e \mu_d E \quad \text{--- (x)}$$

$$\text{again we know, } J = \sigma E \quad \text{--- (xx)}$$

where,

$\sigma \Rightarrow$ conductivity

$E \Rightarrow$ applied electric field

Comparing eqⁿ (x) & (xx) we get,

$$\text{Conductivity } (\sigma) = e N_e \mu_d$$

$$\text{and resistivity } (\rho) = \frac{1}{\sigma} = \frac{1}{e N_e \mu_d}$$

$$\text{Resistance (R)} = \frac{\rho l}{A}$$

Numericals

- ① Electron Mobility is $1400 \text{ cm}^2/\text{Vs}$. Calculate the mean free time of scattering of electrons.
 $m_e^*/m_e = 0.33$.

Solution

we know,

$$\mu = \frac{e\tau}{m_e^*}$$

$$\therefore \tau = \frac{\mu m_e^*}{e}$$

$$\text{Given, } \mu = 1400 \text{ cm}^2/\text{Vs} = 1400 \times 10^{-4} \text{ m}^2/\text{Vs}$$

$$m_e^* = 0.33 m_e = 0.33 \times 9.11 \times 10^{-31} \\ = 3.0063 \times 10^{-31} \text{ kg}$$

$$\therefore \tau = \frac{1400 \times 10^{-4} \times 3.0063 \times 10^{-31}}{1.6 \times 10^{-19}}$$

$$= 2.63 \times 10^{-13} \text{ sec} \quad \underline{\underline{\text{Ans}}}$$

- ② If electrical conductivity of potassium is $1.39 \times 10^5 \text{ S/cm}$, Calculate the drift mobility of electron at room temperature. Molar mass and density of potassium are 39.95 and 0.91 g/cm^3 .

Solution

given,

$$M_{\text{at}} (\text{Atomic mass}) = 39.95 \text{ g/mol} \\ = 39.95 \times 10^{-3} \text{ kg/mol}$$

$$\text{density } (\rho) = 0.91 \text{ g/cm}^3 = 910 \text{ kg/m}^3$$

$$\text{electrical conductivity } (\sigma) = 1.39 \times 10^5 \text{ S/cm}$$

$$\text{Also we know, } \sigma = 1.39 \times 10^7 \text{ S/m}$$

$$\sigma = n e \mu_e$$

$$\text{Free electron concentration } (n) = \frac{\rho \times N_A}{M_{\text{at}}}$$

N_A is avogadro's number which is equal to 6.023×10^{23} and each potassium atom contribute only one free electron thus,

$$n = \frac{910 \times 6.023 \times 10^{23}}{39.95 \times 10^{-3}}$$

$$= 1.37 \times 10^{28} / \text{m}^3$$

$$\text{Now, } \sigma = n e \mu_e$$

$$1.39 \times 10^7 = 1.37 \times 10^{28} \times 1.6 \times 10^{-19} \times \mu_e$$

$$\therefore \mu_e = 6.341 \times 10^{-3} \text{ m}^2 \text{V}^{-1} \text{s}^{-1} \text{ Ans}$$

③ Drift mobility of conduction electron is $43 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and mean speed is $1.2 \times 10^6 \text{ m s}^{-1}$. Calculate the mean free path of electrons between collisions.

Solution

Given,

$$\text{Drift mobility (} \mu_d \text{)} = 43 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1} \\ = 43 \times 10^{-4} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

$$\text{mean free speed (} v \text{)} = 1.2 \times 10^6 \text{ m s}^{-1}$$

$$\text{Now, mean free speed (} v \text{)} = \frac{\text{mean free path (} L \text{)}}{\text{Time (} T \text{)}}$$

$$\therefore L = vT$$

Now, to calculate relaxation time (T) we have,

$$\mu_d = \frac{eT}{m_e}$$

$$\therefore T = \frac{\mu_d m_e}{e}$$

$$T = \frac{43 \times 10^{-4} \times 9.11 \times 10^{-31}}{1.6 \times 10^{-19}}$$

$$= 2.44 \times 10^{-14} \text{ sec}$$

and

$$\text{mean free path (L)} = vT$$

$$= 1.2 \times 10^6 \times 2.44 \times 10^{-14}$$

$$= 2.937 \times 10^{-8} \text{ m}$$

$$= 29.37 \text{ nm} \quad \underline{\text{Ans}}$$

④ Calculate the root mean square velocity of nitrogen molecule in atmospheric air at 27°C . Also calculate the rms velocity in one direction.

Solution

We know, atomic mass of nitrogen atom is $14 \text{ g/mol} = 14 \times 10^{-3} \text{ Kg/mol}$
(Mat)

Also, root mean square velocity $(v_{\text{rms}}) = \sqrt{\frac{3KT}{m}}$

For Nitrogen molecule (N_2), Mass of the molecule $= \frac{2 \times \text{Mat}}{N_A}$

Where, $N_A = \text{Avogadro's number} = 6.023 \times 10^{23} / \text{mol}$

$$\therefore \text{mass of molecule (m)} = \frac{2 \times 14 \times 10^{-3}}{6.023 \times 10^{23}}$$

$$= \frac{\cancel{28} \times 10^{-3}}{6.023 \times 10^{23}}$$

$$4.648 \times 10^{-26}$$

$$\text{Root mean square velocity (V}_{rms}) = \sqrt{\frac{3kT}{m}}$$

$$= \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 300}{4.648 \times 10^{-26}}}$$

$$= 517 \text{ ms}^{-1}$$

Also, rms velocity in one direction

$$(V_{rms})_x = \frac{V_{rms}}{\sqrt{3}} = \frac{517}{\sqrt{3}} = 298 \text{ ms}^{-1}$$

Q.5) Calculate the density of copper with
 $a = 0.362 \text{ nm}$, $M_{at} = 63.5 \text{ g/mol}$, $N_A = 6.023 \times 10^{23} \text{ /mol}$

Solution:

density of copper = $\frac{\text{mass of unit cell}}{\text{Volume of unit cell}}$

$$\text{Volume of unit cell} = a^3$$

Now,

$$\text{mass of unit cell} = \frac{M_{at}}{N_A} \times \text{No. of atoms per unit cell}$$

Copper has FCC structure and thus,
No. of atoms per unit cell = 4.

Now,

$$\begin{aligned} \text{mass of unit cell} &= \frac{63.5 \times 10^{-3}}{6.023 \times 10^{23}} \times 4 \\ &= 4.21 \times 10^{-25} \end{aligned}$$

$$\begin{aligned} \text{density of copper } (\rho) &= \frac{4.21 \times 10^{-25}}{(0.362 \times 10^{-9})^3} \\ &= 8890 \text{ Kg/m}^3 \end{aligned}$$

Ans

Q. ⑥ Calculate the drift mobility of electrons in copper at room temperature. Conductivity, mass and density of copper are $5.9 \times 10^5 \text{ Sm/cm}$, 63.5 g/mol and 8.93 g/cm^3 respectively.

Solution

Given,

$$\text{Atomic mass (Mat)} = 63.5 \text{ g/mol} \\ = 63.5 \times 10^{-3} \text{ kg/mol}$$

$$\text{density } (\rho) = 8.93 \text{ g/cm}^3 = 8930 \text{ kg/m}^3$$

$$\text{Conductivity } (\sigma) = 5.9 \times 10^5 \text{ Sm/cm} \\ = 5.9 \times 10^7 \text{ Sm/m}$$

Also we know, $\sigma = n e l_d$

$$\boxed{\text{or, } l_d = \frac{\sigma}{n e}}$$

to find, the concentration of free electrons (n)
we assume that, ^{one} copper atom contribute 1
free electron to the conduction band.

$$\therefore n = \frac{\rho N_A}{\text{Mat}} \text{ where, } N_A = \text{Avogadro's number} \\ = 6.023 \times 10^{23}$$

$$\therefore n = \frac{8930 \times 6.023 \times 10^{23}}{63.5 \times 10^{-3}}$$

$$\therefore n = 8.47 \times 10^{28} / \text{m}^3$$

$$\text{Now, } l_d = \frac{\sigma}{n e}$$

$$m \mu_d = \frac{5.9 \times 10^7}{8.47 \times 10^{28} \times 1.6 \times 10^{-19}}$$

$$\therefore \mu_d = 4.35 \times 10^{-3} \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$$

$$= 43.53 \text{ cm}^2 \text{V}^{-1} \text{s}^{-1}$$

Ans

Q. 7) What applied electric field can impose a drift velocity equal to 10% of mean speed $1.2 \times 10^6 \text{ ms}^{-1}$ of the conduction electrons in Copper? $\mu_d = 32 \text{ cm}^2 \text{V}^{-1} \text{s}^{-1}$

Solution

Given,

$$\mu_d = 32 \text{ cm}^2 \text{V}^{-1} \text{s}^{-1} = 32 \times 10^{-4} \text{ m}^2 / \text{Vs}$$

$$\text{drift velocity } (V_d) = 10\% \text{ of } 1.2 \times 10^6$$

$$= 1.2 \times 10^5 \text{ m/s}$$

Now, we know,

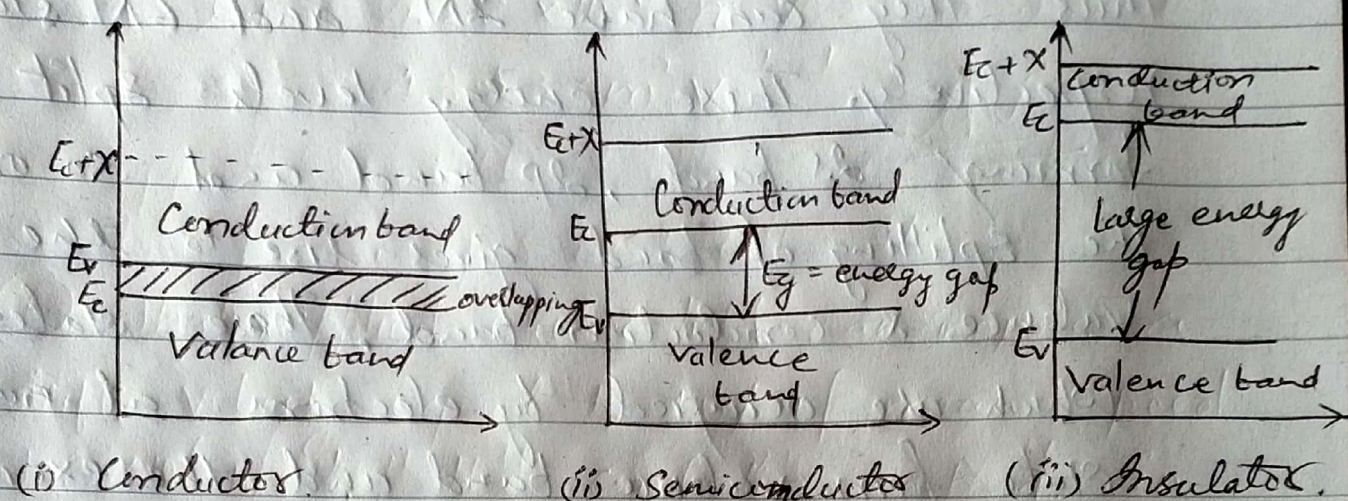
$$V_d = \mu_d E$$

$$\therefore E = \frac{V_d}{\mu_d} = \frac{1.2 \times 10^5}{32 \times 10^{-4}} = 3.75 \times 10^7 \text{ Vm}^{-1} \text{Ans}$$

Band theory of Solids $\Rightarrow$

A useful way to visualize the difference between conductors, insulators and semiconductors is to plot the energies for electrons in the materials. Instead of having discrete energies as in the case of free atoms, the available energy states form bands. Crucial to the conduction process is whether or not there are electrons in the valence band are separated by a large gap from the conduction band, in conductors like metals the valence band overlaps the conduction band and in semiconductors there is a small enough gap between the valence and conduction bands that thermal or other excitation can bridge the gap. With such a small gap, the presence of a ~~small~~ small percentage of a doping material can increase conductivity dramatically.

Energy bands in solids (Conductor, semiconductor and insulator).



* Energy band formation in conductors.

Let us take an example of Li atoms.

The spdf configuration of Li atoms are $1s^2 2s^1$. i.e. The 1s orbital is completely filled and 2s orbital is half filled. The valence electron has energy level of 2s. Consider 'N' Li atoms and because of the close proximity of atoms there is interaction in the energy levels of the electrons and thus the energy level start splitting forming bands.

we know, $2s$ state can have 2 electrons and $2p$ state can have 6 electrons and thus they can have $2N$ and $6N$ states respectively. These energy levels split forming $2s$ band, $2p$ band and so on. The splitting of band gives rise to the valence & conduction band. Those states which are occupied by electrons at normal state form valence band and the excited states form conduction band.

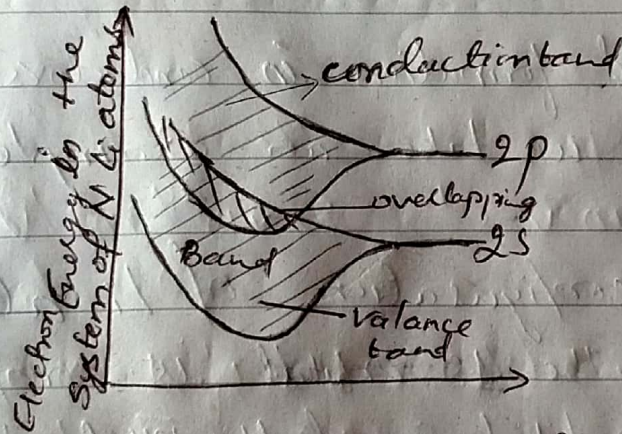


Fig: Band formation in conductors.

* Energy band formation in ~~semiconductors~~ semiconductors.

Let us take an example of Si atoms. The spdf configuration of Si atoms are $1s^2 2s^2 2p^6 3s^2 3p^2$. Here, $3s$ state have 2 electrons

and thus will have $2N$ states (considering N no. of Si atoms in the system) and $3p$ state have 2 electrons and thus will have $2N$ states which are occupied and $4N$ states will be empty. Since $2p$ state can accommodate 6 electrons. Because of the close proximity of Si atoms there is interaction between the neighbouring atoms and thus the energy level start splitting forming valence and conduction band.

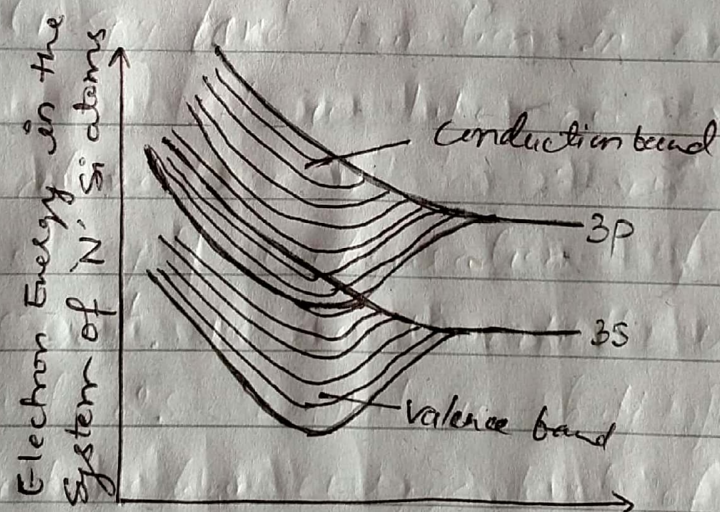


Fig: Band Formation in Semiconductors.